## n - Dimensional Vector Space:

As we know that any point in a plane is determined by its co-ordinates which can be denoted by an ordered pair of real numbers.
Similarly, a point in space can be represented by ordered triads of real numbers, we also have ordered n-tuppleso of real numbers like $(x_1, x_2 \dots x_n)$.
Where $x_1, x_2 \dots x_n$ are real numbers. The set of all n-tupples of real numbers is denoted by $\mathbb{R}^n$.

Vectors:
A set of a numbers $x_1 x_2 \dots x_n$ written in particularly order is called n-dimensional Vector or a vector of order n
The numbers $x_i$ $i = 1, 2, \dots n$ are called components of the Vector. A Vector is denoted by a single letter X or Y etc.
The Vector which represented in rows as $X = (x_1, x_2 \dots x_n)$.
is called row vector and the Vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$ is called Column Vector.

## Linear Combination of Vectors.

A vector $X$ of $n$-dimentional space is called a linear combination of the vectors $X_1 X_2 \ldots X_n$ if there exist numbers (Scalars) $k_1 k_2 \ldots k_n$ such that

$$X = k_1 x_1 + k_2 x_2 \ldots \ldots k_n x_n.$$

## Linear Dependent and Linear Independent.

The vectors $X_1 X_2 \ldots X_n$ are said to be linearly dependent if there exists scalars $k_1 k_2 \ldots k_n$ which are not all zero. Such that $k_1 x_1 + k_2 x_2 \ldots \ldots k_n x_n = 0$

A set of vectors is said to be linearly independent, if it is not linearly dependent. i.e., the vectors $X_1 X_2 \ldots X_n$ are linearly independent, if the relation $k_1 x_1 + k_2 x_2 + k_3 x_3 \ldots k_n x_n = 0$ is satisfied only, when

$$k_1 = k_2 = k_3 = 0.$$

Obs: 1. Let $A$ be a square matrix $|A| = 0$ rows and colums are L.D.

2. Let $A$ be a square matrix $|A| \neq 0$ then rows and columns are L.I.

3. The vector cannot be a member of any linearly independent set. i.e., if a set of vectors includes a zero vectors, it is ~~to be~~ Linearly dependent set.

4 Any subset of L.I set it self L.I

5. If $X_1 X_2 \ldots X_n$ are L.D then any one of them can be expressed as L.C. of the other vectors.

6. If $A$ is of order $m \times n$ and the rank is $r$ then the max. number of L.I. rows or colms of $A$ is $r$.

§ S.t the Vectors (1, 2, 3) (3, −2, 1) (1, −6, 5) form a L.D Set.

Let $X_1 = (1,2,3)$ $X_2 = (3,-2,1)$ $X_3 = (1,-6,5)$.

If these three vectors are L.D we can find the Scalars $k_1 k_2 k_3$ (not all zero). such that the relation

$$k_1 X_1 + k_2 X_2 + k_3 X_3 = 0.$$

$$k_1(1,2,3) + k_2(3,-2,1) + k_3(1,-6,5).$$

i.e., $k_1 + 3k_2 + k_3 = 0$ — (1)

$2k_1 - 2k_2 - 6k_3 = 0$ — (2)

$3k_1 + k_2 + 5k_3 = 0$. — (3).

from 2 and 3.

$$8k_1 - 16k_3 = 0$$

$$k_1 = 2k_3 \quad — (4)$$

Sub 4 in (1)

$$2k_3 + 3k_2 + k_3 = 0$$

$$3k_2 = -3k_3 \Rightarrow k_2 = -k_3 \quad — (5)$$

Sub. 4 and 5 in (1)

We get $2k_3 - 3k_3 + k_3 = 0$

$$k_3 = 0$$

For any value of $k_3$ eqn 1 is satisfied.

if we take $k_3 = 1$. then $k_2 = -1$ $k_1 = 2$.

Hence $k_1 = 2$, $k_2 = -1$, $k_3 = 1$.

∴ the given vectors are L.D. and their relationship is

$$2X_1 - X_2 + X_3 = 0.$$

Method 2

The vectors $X_1 X_2 X_3$ from a square matrics.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 1 & -6 & 5 \end{bmatrix} \Rightarrow |A| = 0.$$

$\cdot\ 1(0+6) - 2(-15-1) + 3(-18+2) = 16+32-48 = 0$

$\therefore$ The given vectors are L.D since $|A| = 0$.

Method 3

Writing $X_1 X_2 X_3$ as vectors, we get

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 1 & -6 & -5 \end{bmatrix}$$

Applying $R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - R_1$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -8 & -8 \\ 0 & -8 & -8 \end{bmatrix}$$

Applying $R_3 \to R_3 \to R_3 - R_2$, $R_2/-8$

In the echelon form of the matrix, the no. of nonzero (L.I). Vectors $= 2$ (< the no. of given vectors).

$\therefore$ $X_1 X_2, X_3$ are L.D.

S.t that the vectors $X_1 = (2,-2,1)$ $X_2 = (1,4,-1)$ and $X_3 (4,6,-3)$ are linearly independent.

Sol. Consider the L.C of vectors.

$$k_1X_1 + k_2X_2 + k_3X_3 = 0.$$

i.e., $k_1(2,-2,1) + k_2(1,4,-1) + k_3(4,6,-3)$.

$$2k_1 + k_2 + 4k_3 = 0. \quad \text{---} (1)$$
$$-2k_1 + 4k_2 + 6k_3 = 0 \quad \text{---} (2)$$
$$k_1 - k_2 - 3k_3 = 0. \quad \text{---} (3).$$

from 1 & 2.

$$k_2 + 2k_3 = 0.$$

from 2 & 3. $k_2 = 0$.

$\therefore k_1 = 0, k_3 = 0.$

$\therefore$ The given vectors are L.I.

Sol. Here the vectors $X_1 X_2 X_3$ form a square matrix

$$A = \begin{bmatrix} 2 & -2 & 1 \\ 1 & 4 & -1 \\ 4 & 6 & -3 \end{bmatrix}$$

$$|A| = -20 \neq 0.$$

Also rank of the matrix from echelon form is $= 3 =$ No. of given vectors.

$X_1 X_2 X_3$ are L.I. vectors

Definition of Basis:

Let $V$ be a vector space over the field $F$. A subset $B = \{v_1, v_2 \ldots v_n\} \subseteq V$ is called the basis of $V$ if

(i) $B$ is L.I

(ii) $B$ spans $V$ (i.e., every vector in $V$ can be expressed as a L.C of vectors in $B$).

$L(B) = V$.

Dimension: The dimension of a vector space $V$, denoted by $\dim(V)$, the no. of vectors in a basis of vector space $V$. (or) The no. of elements in a basis is called dimension of vector space $V$.

Prob: Check whether the set $B = \{(1,0), (0,1)\}$ is a basis of vector space $\mathbb{R}^2$ or not?

Sol: If $B$ is the basis of vector space $\mathbb{R}^2$ then it satisfies following conditions. i.e.,

(i) the set of vectors in $B$ L.I. and

(ii) $B$ spans $V$ i.e, any vector space in $V$ can be expressed as L.C. of vectors $B$.

First we check the first condition

i.e., $\alpha(1,0) + \beta(0,1) = (0,0)$

$\Rightarrow \alpha = 0, \beta = 0$.

$\therefore$ The vectors are L.I in B.

Second condition

Let $(x,y) \in \mathbb{R}^2$ then $(x,y)$ can be expressed as L.C of vectors of B i.e.,

$(x,y) = x(1,0) + y(0,1)$.

$\therefore$ $B = \{(1,0)\ (0,1)\}$ is the basis of vector space $\mathbb{R}^2$.

The dim. of $\mathbb{R}^2 = 2$.

Ex: 2. Check whether the set $B = \{(1,0,0), (0,1,0), (0,0,1)\}$ is a basis of vector space $\mathbb{R}^3$ or not.

Let L.C $\alpha(1,0,0) + \beta(0,1,0) + \gamma(0,0,1) = (0,0,0)$.

$\alpha = 0, \beta = 0, \gamma = 0$.

$\Rightarrow$ B is L.I.

and also $(x,y,z) \in \mathbb{R}^3$ i.e.

$(x,y,z)$ can be written as L.C of set of vectors in B.

$(x,y,z) = x(1,0,0) + y(0,1,0) + z(0,0,1)$.

$\Rightarrow$ B is the Basis of the vector space $\mathbb{R}^3$.

$\dim B = 3$.

Prob: Find a basis of the subspace $\mathbb{R}^3$ defined by

$W = \{(x,y,z) \in \mathbb{R}^3 \mid x+y+z=0\}$.

given that $W$ is a subspace of $\mathbb{R}^3$ now

$x+y+z=0$ can be written as

$$z = -x-y$$

So, every vector is of the vector form ie

$$(x, y, -x-y) = (x,y,z) \in \mathbb{R}^3$$

$$= x(1,0,-1) + y(0,1,-1)$$

the set of B has only two vectors $(1,0,-1)$ and $(0,1,-1)$ which is L.I.

So $B = \{(1,0,-1), (0,1,-1)\}$ is the basis of the given subspace $W$.

$\therefore$ dim of $W = 2$

Prob: Find basis of the column space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

The given matrix has column vector set

$B = \{(1,2,1), (2,4,2), (3,6,3)\}$ first we check the given vector of column space are L.I. or not.

Clearly, the set of vector B is L.D

because the vector $(1,2,1)$ is multiple of other two vector

hence, the basis of column space $B = \{(1,2,1)\}$.

The dim of $B = 1$.

Prob. Find a basis of the row space of $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

Soln: Given $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ Convert B into row reduce

apply $-4R_1 \to R_2$ We have $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \end{bmatrix}$

$\sim R_2/-3 \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix}$

i.e., row space has two vectors $B = \{(1,2,3), (0,1,2)\}$.

B is L.I. and also span B is in V.

$B = \{(1,2,3), (0,1,2)\}$ is a basis of the row space of given matrix $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

dim $B = 2$.

Prob: Find a basis for the solution space of

$$x_1 + 2x_2 + x_3 = 0$$

Soln: The given eqn. $x_1 + 2x_2 + x_3 = 0$ can be written as $x_1 = -2x_2 - x_3$

i.e., no. of independent variable = no. of variable − (no. of eqn)

$$= 3 - 1 = 2.$$

∴ We have $x_2, x_3$ are independent variable.

So, any vector $(-2x_2 - x_3, x_2, x_3)$ is the linear combination of $(-2x_2 - x_3, x_2, x_3) = x_2(-2,1,0) + x_3(-1,0,1)$.

$\Rightarrow B = \{(-2,1,0), (-1,0,1)\}$ is a basis for the solution space of $x_1 + 2x_2 + x_3 = 0$ dim $B = 2$.

Prob: Find a basis of the subspace spanned by $\{(1,2,3), (2,4,6), (1,1,1)\}$.

Soln: The vector $(2,4,6)$ is dependent (since multiple of $(1,2,3)$) but other two vectors $(1,2,3)$ and $(1,1,1)$ are L.I.

∴ $B = \{(1,2,3), (1,1,1)\}$ is a basis of the subspace spanned by the set $\{(1,2,3), (2,4,6), (1,1,1)\}$